EECS 4422 Computer Vision

Frequency-Domain Denoising

Thien Nguyen - 219183821 - thienyu@my.yorku.ca

Implementing and comparing Fourier-domain restoration methods against spatial-domain baselines for denoising

Image source

Clean reference images are taken from CBSD68 (68 colour natural images, 321×481), obtained from the benchmark collection at https://github.com/majedelhelou/denoising_datasets.

Degradation model — Synthetically noising the input images

Following the course notation, g denotes the original image and f the degraded observation. Unlike the deblurring half there is no blur kernel ($H = 1$):

$$f(x, y) = g(x, y) + n(x, y) \tag{1}$$
$$F(u, v) = G(u, v) + N(u, v) \tag{2}$$

g = original (clean) image, n = additive white Gaussian noise of known standard deviation σ, f = observed noisy image; capitals denote the corresponding 2-D discrete Fourier transforms, and $\hat{G}$ denotes the restored estimate of G.

Noise is synthesised in floating-point intensity units with $\sigma = 25/255$, then written to 8-bit PNG (so a small quantisation term is also present when restoring from disk).

Ideal low-pass filter

Noise is broadband while natural-image energy concentrates at low frequencies, so a first restoration keeps only a disk of radius D_0 in the Fourier plane:

$$H(u, v) = \begin{cases} 1, & D(u, v) \leq D_0, \\ 0, & D(u, v) > D_0, \end{cases} \qquad \hat{G} = H\,F, \qquad D_0 = 0.08 \tag{3}$$

where $D(u, v)$ is the distance from DC in cycles per sample. The brick-wall cutoff produces Gibbs ringing around edges.

Steps

1. Build the radial distance grid $D(u, v)$ with DC at the spectrum centre.
2. Form $H = [D \leq D_0]$.
3. For each colour channel: $F = \mathcal{F}\{f\}$, $\hat{G} = HF$, $\hat{g} = \mathrm{Re}\ \mathcal{F}^{-1}\{\hat{G}\}$.

Butterworth and Gaussian low-pass filters

A Butterworth filter of order n softens the cutoff:

$$H(u, v) = \frac{1}{1 + \big(D(u, v)/D_0\big)^{2n}}, \qquad n = 2,\ D_0 = 0.08 \tag{4}$$

A Gaussian low-pass has no ringing at all:

$$H(u, v) = \exp\left(-\frac{D(u, v)^2}{2D_0^2}\right) \tag{5}$$

Both are applied exactly as the ideal filter: multiply F by H, then invert. They are the Fourier-domain counterparts of ordinary smoothing, and give a controlled comparison against the spatial Gaussian baseline below.

Wiener filter ($H = 1$)

With no blur the Wiener solution is a per-frequency shrink by the noise-to-signal power ratio:

$$\hat{G}(u,v) = \frac{P_g(u,v)}{P_g(u,v) + P_n(u,v)}\, F(u,v), \qquad P_n = MN\sigma^2 \tag{6}$$

P_g is unknown, so the practical filter estimates it from the noisy periodogram, $P_g \approx \max(|F|^2 - P_n,\ 0)$. We also report an oracle that uses the true $|G|^2$; that score is an upper bound, not an achievable method (same role as in the deblurring report).

Spatial-domain baselines

Three pixel-domain denoisers are run on the same noisy inputs:

- Gaussian smoothing — isotropic blur with $\sigma_s = 1$ px.
- Median filter — 3×3 disk median per channel.
- Bilateral filter — spatial $\sigma_s = 2$ px and tonal $\sigma_c = 0.08$, so edges are preserved better than a plain Gaussian.

These are the spatial baselines the project brief asks for: the Fourier methods above are compared against them on PSNR, SSIM and runtime.

Results

All methods are run over the same 68 CBSD68 images, corrupted with $\sigma = 25/255$ Gaussian noise and written to 8-bit PNG. Each restored image is clipped to $[0, 1]$ before scoring. Blue bars in the chart are Fourier-domain methods; green bars are spatial baselines; the hatched bar is the oracle Wiener.

Method	PSNR (dB)	Δ PSNR	SSIM	ms/image
Noisy input	20.54	—	0.4174	—
Ideal LPF	22.71	+2.17	0.5312	13
Butterworth LPF	23.43	+2.90	0.6055	14
Gaussian LPF	23.87	+3.33	0.6341	14
Wiener filter	24.26	+3.72	0.5512	14
Wiener filter (oracle) *	27.36	+6.82	0.7431	20
Gaussian (spatial)	25.96	+5.43	0.6913	2
Median filter	24.28	+3.75	0.5552	12
Bilateral filter	25.94	+5.41	0.6847	105

Table 1. Mean PSNR and SSIM over 68 images, $\sigma = 25/255$. Asterisk marks the oracle (uses ground-truth P_g).

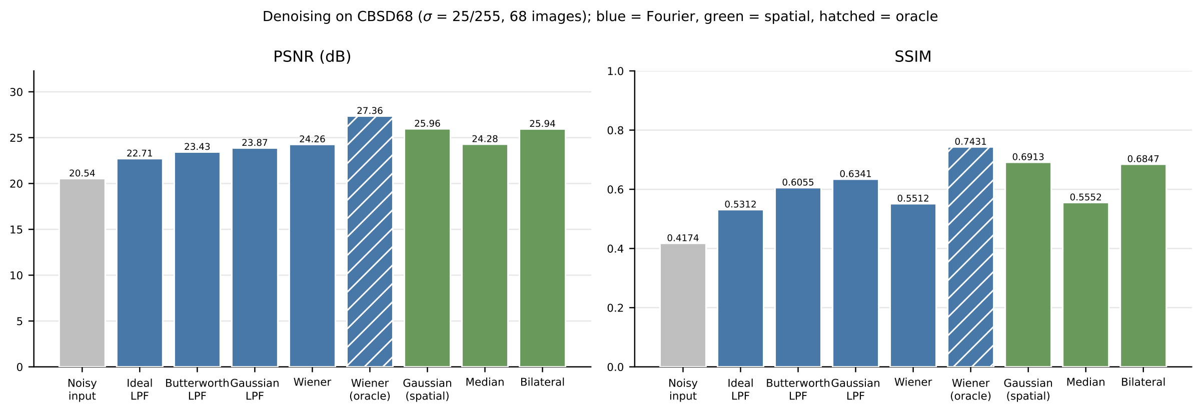
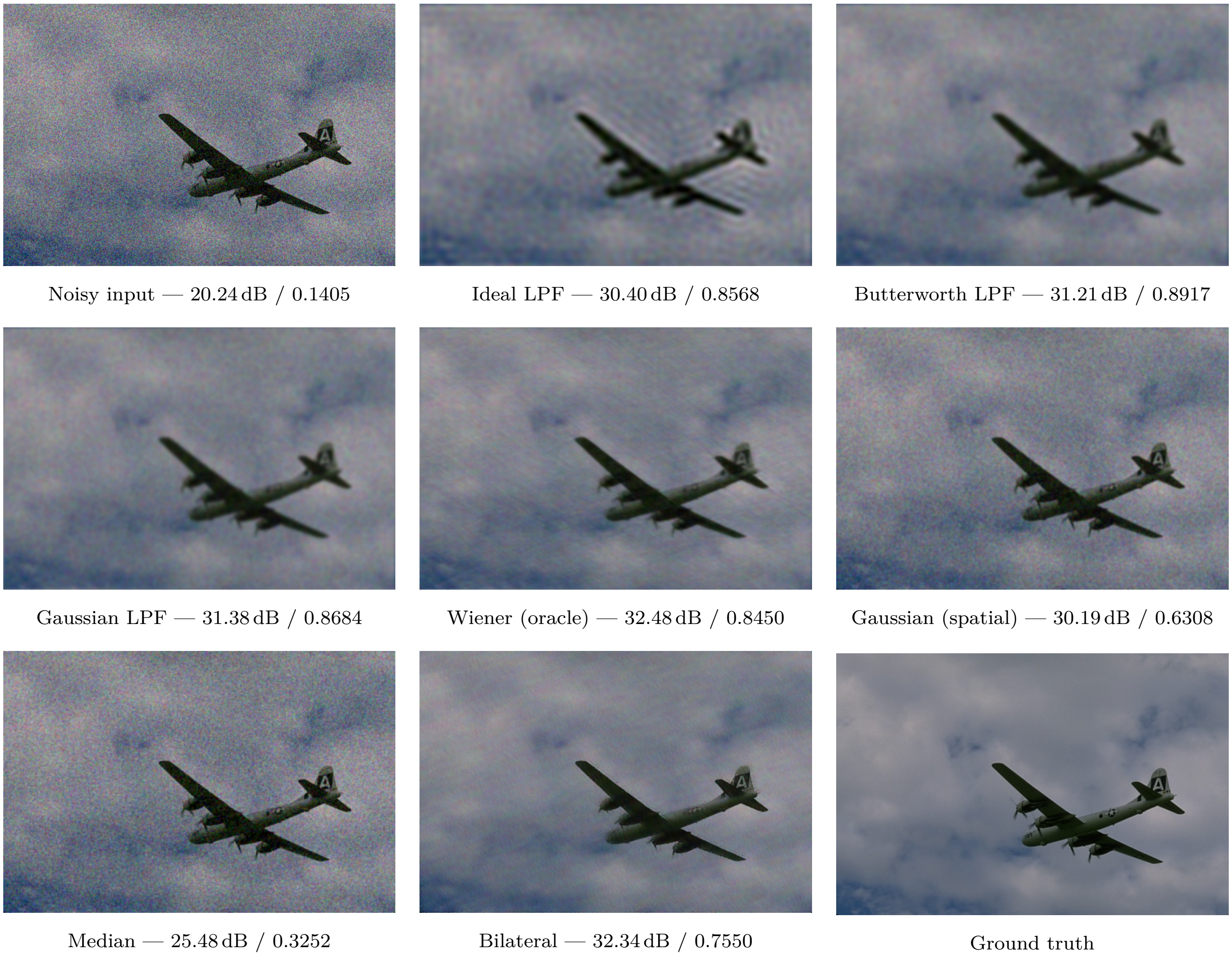


Figure 1. The same figures. Hatched = oracle Wiener.

Sample restorations

One grid per image, each cell captioned with its own PSNR and SSIM against the ground truth. Generated by save_panels(), so the scores below cannot drift from those in the table.



Noisy input — 20.24 dB / 0.1405

Ideal LPF — 30.40 dB / 0.8568

Butterworth LPF — 31.21 dB / 0.8917

Gaussian LPF — 31.38 dB / 0.8684

Wiener (oracle) — 32.48 dB / 0.8450

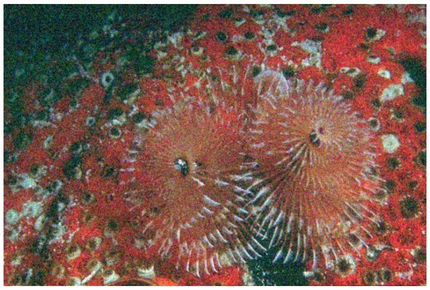
Gaussian (spatial) — 30.19 dB / 0.6308

Median — 25.48 dB / 0.3252

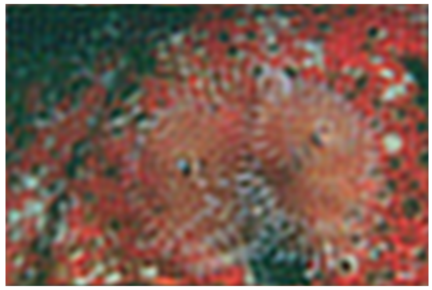
Bilateral — 32.34 dB / 0.7550

Ground truth

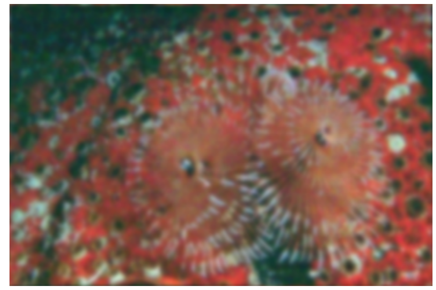
Figure 2. 3096.png restored from $\sigma = 25/255$ noise. Read left to right, top to bottom.



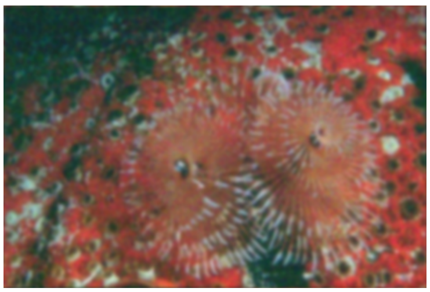
Noisy input — 20.27 dB / 0.4686



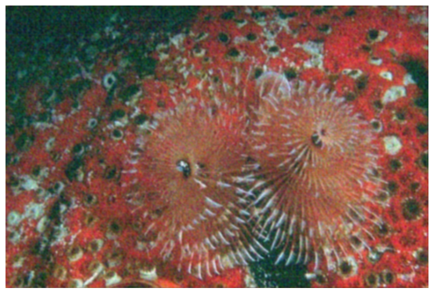
Ideal LPF — 22.44 dB / 0.4736



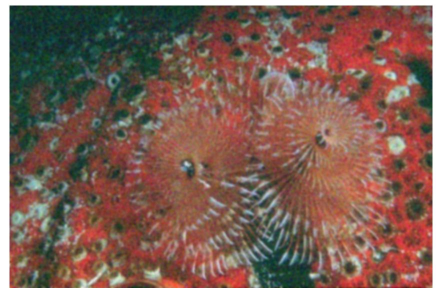
Butterworth LPF — 23.45 dB / 0.5681



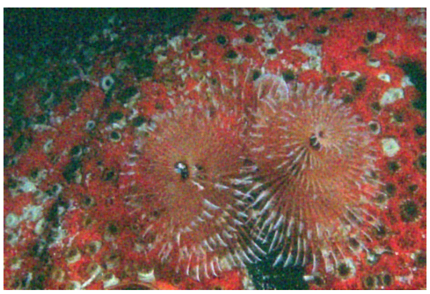
Gaussian LPF — 24.13 dB / 0.6219



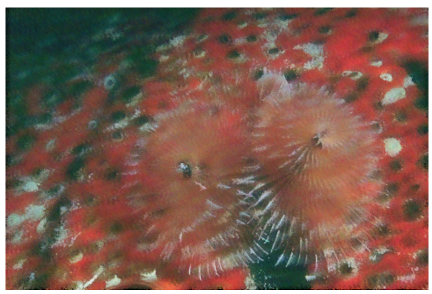
Wiener (oracle) — 27.67 dB / 0.7801



Gaussian (spatial) — 26.89 dB / 0.7535



Median — 24.57 dB / 0.6350



Bilateral — 23.94 dB / 0.5970

Ground truth

Figure 3. 12084.png restored from $\sigma = 25/255$ noise. Read left to right, top to bottom.

Noisy input — 20.39 dB / 0.4386

Ideal LPF — 23.18 dB / 0.5329

Butterworth LPF — 23.92 dB / 0.6016

Gaussian LPF — 24.38 dB / 0.6350

Wiener (oracle) — 27.44 dB / 0.7519

Gaussian (spatial) — 26.63 dB / 0.7195

Median — 24.44 dB / 0.5882

Bilateral — 25.51 dB / 0.6680

Ground truth

Figure 4. 16077.png restored from $\sigma = 25/255$ noise. Read left to right, top to bottom.

Noisy input — 20.42 dB / 0.4334

Ideal LPF — 22.38 dB / 0.5190

Butterworth LPF — 23.17 dB / 0.5971

Gaussian LPF — 23.68 dB / 0.6295

Wiener (oracle) — 26.96 dB / 0.7255

Gaussian (spatial) — 25.99 dB / 0.7036

Median — 24.28 dB / 0.5710

Bilateral — 25.18 dB / 0.6551

Ground truth

Figure 5. 19021.png restored from $\sigma = 25/255$ noise. Read left to right, top to bottom.

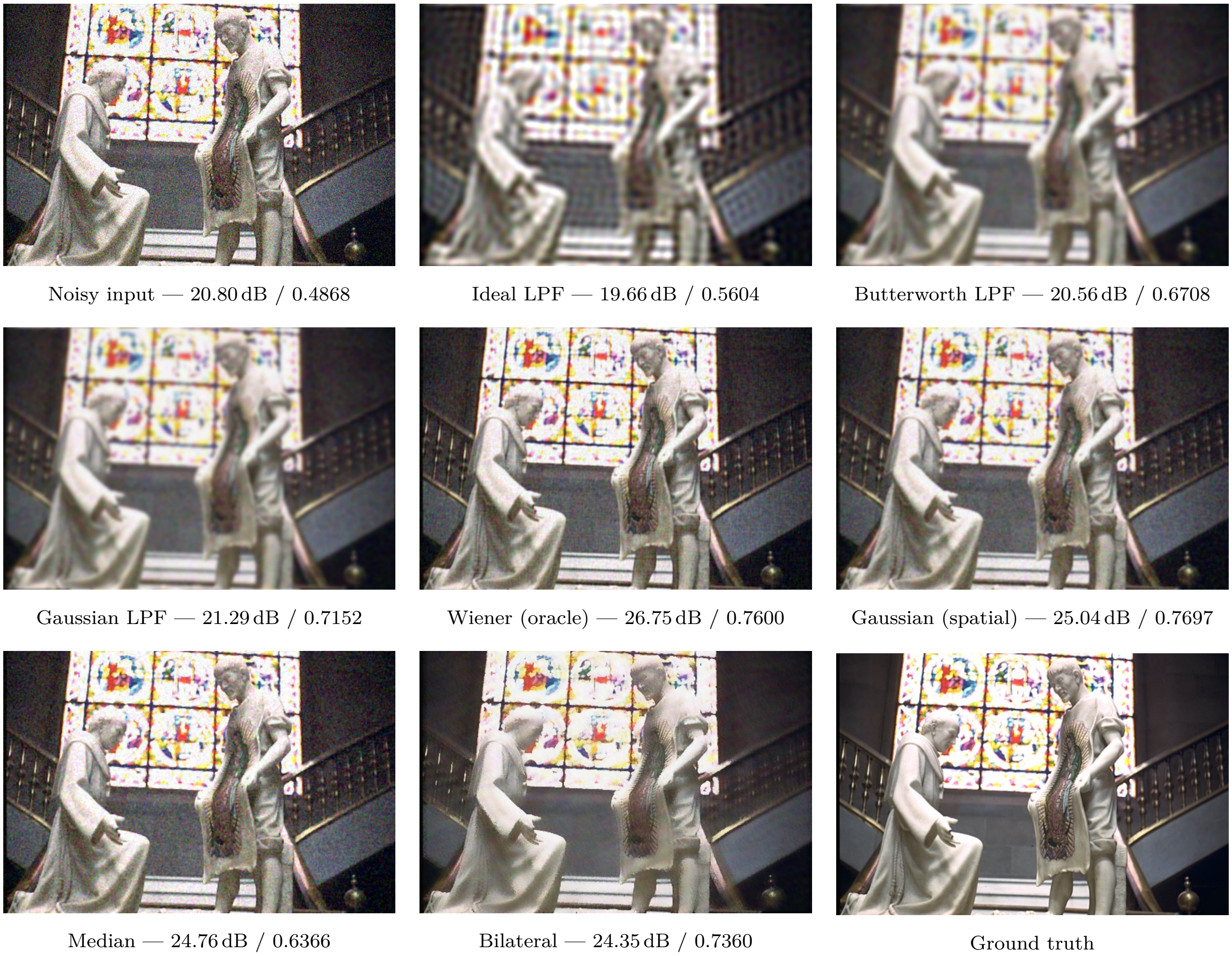


Figure 6. 24077.png restored from $\sigma = 25/255$ noise. Read left to right, top to bottom.